

Agent In The Loop

Round I

IDEA SUBMISSION

HACKFEST_{2.0}

Team Details:

S no.	Team Member Name	Team Member Role	Team Member Mail ID
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Track Chosen:

Intelligent Data Dictionary Agent



Problem Description:

Enterprises rely on complex databases, yet their documentation is often outdated, fragmented, and purely technical—creating a critical gap between data availability and data understanding. This leads to poor data trust, slow decision-making, and inefficiencies.

There is a need for an AI-powered platform that automatically extracts metadata, evaluates data quality, generates business-friendly documentation, and enables intuitive natural language exploration of enterprise data assets.



Proposed Solution / Idea Overview:

We propose an AI-powered Data Documentation & Discovery Platform that transforms complex enterprise databases into business-ready knowledge systems. It connects to multiple databases, automatically extracts schema metadata, performs real-time data quality analysis, and generates AI-driven business summaries. Through statistical profiling and contextual insights, it converts technical structures into clear, trustworthy documentation. A conversational interface enables natural language exploration, schema understanding, and SQL generation. Built with a modular, scalable architecture, it supports incremental updates, visualization, and multi-format exports—eliminating manual documentation, improving data governance, and accelerating confident, data-driven decision-making.





- ♦ **Intelligent Metadata Extraction**

Automated multi-database schema discovery with relationship mapping and change detection.



- ♦ **Real Time Data Quality Intelligence**

Advanced statistical profiling with completeness, freshness, key health, and anomaly detection.



- ♦ **AI-Generated Business Context**

Business-friendly summaries, usage insights, and SQL recommendations powered by AI.



- ♦ **Conversational Chatbot Feature**

Natural language chat interface for intuitive schema exploration and query generation.

THANK YOU !!

From Raw Schema to Real Intelligence.

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